

---

## SECTION A: MULTIPLE CHOICE QUESTIONS (MCQs)

- Q1.** Which of the following is a fundamental concept describing the operation of any system? [1]
- a) Its age and size
  - b) Its objective or purpose
  - c) Its financial cost
  - d) Its isolation from factors
- Q2.** In the standard ASCII text encoding scheme, how many bits are used to represent a single character? [1]
- a) 7 bits
  - b) 8 bits
  - c) 16 bits
  - d) 32 bits
- Q3.** Which basic logical Boolean law is represented by the mathematical expression  $A + 1 = 1$ ? [1]
- a) Identity Law
  - b) Idempotent Law
  - c) Null Law
  - d) Complement Law
- Q4.** What is the very first step in the systematic, six-step process of troubleshooting computer systems? [1]
- a) Establish a Theory of Probable Cause
  - b) Verify Full System Functionality
  - c) Document Findings, Actions, and Outcomes
  - d) Identify the Problem
- Q5.** Which layer of the OSI networking model is responsible for routing data packets between different networks? [1]
- a) Physical Layer
  - b) Data Link Layer
  - c) Network Layer
  - d) Transport Layer
- Q6.** Which transmission mode allows data communication in both directions, but not at the same time? [1]
- a) Simplex
  - b) Half-Duplex
  - c) Full-Duplex
  - d) Multiplexing

## SECTION B: SUBJECTIVE — SHORT QUESTIONS

- Q1.** Perform binary arithmetic addition on the following representations:  $(1100)_2 + (1011)_2$ . Show your final answer. [2]
- Q2.** State the exact technical purpose of Analog-to-Digital Converter (ADC) and Digital-to-Analog Converter (DAC). [2]
- Q3.** Construct a complete truth table for two binary variables (A, B) utilizing the logical AND operation. [2]
- Q4.** Enlist all six steps of systematic computer troubleshooting in their correct sequential order. [2]
- Q5.** Define data communication and state its five fundamental components. [2]
- Q6.** Explain the physical difference between Star topology and Bus topology in a computer network. [2]
- Q7.** Differentiate between Half-Duplex and Full-Duplex transmission modes, providing a daily-life example of each. [2]
- Q8.** What is the specific role of the Network Layer in the 7-layer OSI Model? [2]
- Q9.** Briefly explain the primary purpose of the Domain Name System (DNS) in computer networks. [2]

## SECTION C: SUBJECTIVE — DETAILED/LONG QUESTIONS

- Q1.** Explain the Von Neumann Architecture of a computer. [8]
- Q2.** Discuss Logical Operations and Logic Gates in detail. [8]  
(Define, Working, Truth table, Diagram of NOT, OR, AND, NAND, XOR)